Orthodontics & Craniofacial Research

WILEY

RESEARCH ARTICLE OPEN ACCESS

The Mechanical Properties of Orthodontic Aligners of Clear Aligner After Intraoral Use in Different Time Periods

Kalliopi Siotou[1] | Theofanis Chountalas[2] | Athanasios Katsavrias[3] | Charalampos Siotos[4] | Konstantinos Mpalias[3] | Dionysios Semitekolos[3] | Constantinos Charitidis[3] | Apostolos I. Tsolakis[1]

[1]Department of Orthodontics, National and Kapodistrian University of Athens, Athens, Greece | [2]Department of Laboratory of Heterogeneous Mixtures & Combustion Systems, Mechanical Engineering School of the National Technical University of Athens, Athens, Greece | [3]Research Lab of Advanced, Composite, Nano Materials & Nanotechnology, Chemical Engineering School of the National Technical University of Athens, Zografou, Greece | [4]Division of Plastic and Reconstructive Surgery, Rush University Medical Center, Chicago, Illinois, USA

Correspondence: Kalliopi Siotou (kalliopisiotou@gmail.com)

Received: 11 November 2023 | **Revised:** 28 September 2024 | **Accepted:** 4 October 2024

Funding: The authors received no specific funding for this work.

Keywords: ageing | clear aligner | intraoral | mechanical properties | orthodontic aligners

ABSTRACT

Objective: Although the technique of orthodontic aligners has risen in popularity, their mechanical properties have not been thoroughly investigated. The aim of this study was to evaluate the mechanical properties of the orthodontic aligners Clear Aligner after intraoral use for 7, 10 and 14 days, and to compare them with as-received aligners (0 days). It was also sought to examine the properties of the unprocessed raw material (polyethylene glycol terephthalate) used to manufacture these aligners.

Materials/Methods: Thirty-two aligners by four patients were evaluated and studied at 0, 7, 10, 14 days of use. Each aligner was divided into three segments (two posterior and one anterior), which resulted in 96 samples. Also, 16 samples of unprocessed material were studied. For all samples, elastic modulus, ultimate tensile stress (UTS) and yield stress were calculated by conducting tensile testing. Additionally, material hardness was tested. The two-tailed Mann–Whitney test was performed, having set the level of significance at $p = 0.05$.

Results: Analysis of the measurements indicated a statistically significant decrease in elastic modulus between days 0 and 14 of use, of UTS between days 0 and 7, 7 and 10, and of yield stress between days 0 and 7. For hardness, in every period, posterior segments demonstrated significantly higher values than anterior segments. All properties of the unprocessed material were statistically significantly higher than the processed samples.

Conclusions: The unprocessed material presented significant differences in every property tested in comparison to the processed aligners. The processed material showed further deterioration over time during use. The present study provides evidence that thermoforming and ageing affect the mechanical properties of the aligners.

1 | Introduction

The use of orthodontic aligners to achieve the desired tooth movement has become increasingly popular over the years. Numerous manufactures and techniques have emerged, offering a variety of options, such as Invisalign, Clear Aligner, K-Line, Duran or in-house three-dimensional (3D) printed aligners.

In comparison with conventional fixed appliances, orthodontic aligners have the advantage of decreased chair time, better oral hygiene condition, gingival health, less pain and more concealed treatment [1–3]. However, many researchers argue that aligners may result to less predictable and accurate orthodontic outcomes, as specific orthodontic movements, such as extrusion and rotation, are difficult to be accomplished and more than half of the cases may require refinement [4, 5].

Polyester, polyurethane and polypropylene are the predominant thermoplastic materials in polymer blends used to manufacture orthodontic aligners. Polyurethane is one of the most versatile thermoplastic engineering compounds, with excellent physical properties, chemical and abrasion resistance, and ease of processing. However, it is not an inert material, and its properties may be affected by heat, moisture or contact with salivary enzymes [6, 7]. Invisalign aligners are made of a polyurethane-based material [8, 9], whereas Clear Aligner by polyethylene glycol terephthalate (PETG) [9].

Fang et al. [10] evaluated the Invisalign thermoformed aligner material condition before and after clinical application, including its mechanical properties, surface morphology, internal structure, and chemical composition and concluded that the mechanical properties of aligners were not significantly affected by the oral environment. On the other hand, Bradley et al. [8] performed similar research and identified significant changes in the mechanical properties of the aligners after extended oral use.

Can et al. [11] studied the alterations of the mechanical properties of in-house three-dimensional (3D) printed orthodontic aligners and concluded that the mechanical properties of the aligners were not affected after 1 week of intraoral use. In 2019, Jindal et al. [12] found that the 3D printed aligners could deform elastically and reversibly for lower displacements, whereas larger displacements result to plastic, hence irreversible, deformation. According to the authors, the ideal aligner should exert continuous light force, its material should be stiff enough to withstand the applied force, and the stress relaxation curve should be quite flat.

Despite the increasing interest in orthodontic aligner treatment, the mechanical properties of clinically worn thermoformed aligners have not been thoroughly investigated [10] and more studies are needed to evaluate the effect of ageing on their mechanical properties [13, 14].

The aim of this study was to investigate the tensile properties and hardness of Clear Aligner orthodontic aligners (Scheu-Dental, Iserlohn, Germany), after their intraoral use for 7, 10 and 14 days and to compare them with unused aligners (as-received [0 days]), and their unprocessed material (PETG). The null hypothesis tested is that intraoral ageing of the aligners does not adversely affect the tensile properties and hardness of the aligners.

2 | Materials and Methods

2.1 | Patient Population

After obtaining approval from the Ethic approval committee of the National and Kapodistrian University of Athens, we prospectively enrolled in the study patients presented at the Department of Orthodontics of the National and Kapodistrian University of Athens (Athens, Greece). We considered eligible all patients who were seeking initial full orthodontic treatment with Clear Aligner on both arches, had full dental arches [7] with absence of transverse and vertical occlusal discrepancies, had no periodontal disease, parafunctional habits or other systemic diseases or syndromes, and were willing to provide informed consent and follow the provided instructions. We excluded patients who had extracted or missing teeth, other than the third molars, had to use transmaxillary elastics, and whose aligners presented signs of abuse, such as cracks or breaks. We did not apply any exclusion criteria on the basis of age, race or gender.

The exclusion criteria were applied on the basis of the results of previous studies which have shown that the aforementioned factors could affect the mastication ability and create uneven distribution of forces between the patients. Particularly, temporomandibular disorder (TMD), as also sleep bruxism, can alter the masticatory muscles' functions [15, 16] and eventually cause longer acting and greater occlusal forces to be exerted on the aligners [17, 18]. Periodontitis and extractions also affect the mastication force causing a negative impact on masticatory cycle efficiency and molar bite force, according to the residual number of occlusal support areas [19, 20]. Patients using transmaxillary elastics were excluded from our sample to eliminate any possible influence elastics have on jaw movement [21]. Last, patients with systematic diseases that affect mastication, such as diabetes, stroke, chronic obstructive pulmonary disease and Sjögren's syndrome [22–24], were excluded.

Patients who agreed to participate were instructed to use the aligners for at least 22 h a day, and to remove them only during mastication, liquid consumption and oral hygiene care. They were also requested to use a diary to record the daily use of the aligners to confirm the exact duration of their use. Participants were asked to use the first set of aligners for 7 days, the second one for 10 days and the third for 14 days. They were then asked to return them to their orthodontist. An unused set (as-received) from each patient was taken for analysis.

2.2 | Sample Preparation

In our study, aligners of similar thicknesses (0.75 mm) were used, as Kwon et al. [25] concluded that the forces delivered by aligners are influenced by the type and the thickness of the material. The importance of the aligners' thickness is correlated with the amount of load exerted on the periodontal ligament and with the orthodontic force delivered to achieve the planned sequential dental movements [26]. Appliances fabricated from thicker material (0.75 or 0.8 mm), according to Kohda et al. [14] produce significantly greater force than those fabricated from thinner material (0.4 or 0.5 mm).

Following retrieval, the aligners underwent ultrasound cleaning, and each aligner was divided into three different sections, two posterior and one anterior. Specifically, the posterior segments spanned from the second molar to the first premolar, and the anterior included the canine-to-canine section. The right and left segments of the aligners were studied together and are referred as posterior. From each segment a specimen was made that included the occlusal surface for the posterior segments, and the buccal surfaces for the anterior.

The segments were prepared using a sharp tool with straight edges to lower the possibility of shear stress microcracks forming. Each segment was measured in the three dimensions of length, width and thickness. In collaboration with the Chemical Engineering School of the National Technical University of Athens (Athens, Greece), the different samples of the orthodontic aligners were subjected to a study of their tensile properties and hardness. Separate specimens from the unprocessed material of Clear Aligner were also submitted to tensile testing. Ten specimens were prepared using a water jet machine and six with the same sharp tool as the one used for the aligners, both according to ISO 6892-1:2019 specifications (Figure S1) [27].

2.3 | Assessment of Mechanical Properties and Analysis

Tensile tests were conducted to assess the mechanical properties of the samples, specifically elastic modulus (E), ultimate tensile stress (UTS) and yield stress (YS). Elastic modulus provides the correlation of applied force to resulting deformation and is defined as the ratio of stress to strain. Materials with low elastic modulus tend to be less brittle than materials with high elastic modulus, and to exert lower elastic forces [28, 29]. Yield stress is the point at which a predetermined amount of permanent deformation occurs, and a high value indicates better resistance to deactivation caused by plastic deformation, which favours applications relying on elastic forces. Ultimate tensile stress is the maximum stress that a material can withstand, without breaking, while being stretched or pulled and signifies the probability of a material fracturing under load [28]. All mechanical properties were measured according to international standard specifications (ISO 14577-2016) [30]. Elastic modulus and tensile stress were tested with a SAUTER TVO 500N500S tensile machine by Sauter (GmbH, Wutöschingen, Germany, error = ±2%) with a load cell of 500 N and 15 mm/min displacement speed. As sliding and minor strain during the placing of the segments at the machine grips contribute to measurement error, we aimed towards accurate placement with minimum application of force. For the hardness measurement a Durometer Type D by Instron (Boston, Massachusetts, USA, error = ±2%) was used, according to the ASTM D2240 standard [31]. Specific anterior and posterior points were measured for hardness, prior to tensile properties measurements, and in particular the buccal surface of the central incisors and the mesial buccal cusp of the first molars. Samples with identified flaws were excluded from the tensile analysis.

From each measurement, a stress–strain diagram was obtained and studied. The elastic modulus was defined as the slope of the obtained linear fit regression graph of each diagram. From each diagram a regression, graph was manually created and through parallel shifting along the x-axis until intersection with the stress–strain curve at 0.1% relative strain, the 0.1% offset yield strength was defined. Ultimate tensile stress was the highest stress value of each diagram (Figure S2). To estimate the required sample size, we conducted a power analysis. Based on the previous literature, the minimum sample size to detect an effective size of 0.5 for all the properties tested, setting alpha to 0.05 and power to 0.80 was at least 15 segments of aligners for each period. The results from the different samples were compared using the two-tailed Mann–Whitney test. For all the statistical tests, the level of significance was set at 95% ($p = 0.05$).

3 | Results

The study sample consisted of four patients treated with Clear Aligner aligners. Each patient provided four sets of aligners and each aligner was divided into three segments, totalling to 96 samples that were tested for their mechanical properties. In addition, 16 samples of unprocessed material were included (six prepared by a sharp tool and 10 by a water jet machine). It is noted that 16.96% of samples had to be excluded from the analysis because they presented microcracks or flaws at the stage of preparation. The summary of the measurements is found in Tables 1, Table S1 (for the processed material) and Table 2 (for the unprocessed material).

Based on our findings, the average elastic modulus of the segments (Figure 1) presented statistically significant reduction from day 0 of use (911.66 ± 139.90 MPa) to day 14 (746.41 ± 145.41 MPa, $p = 0.038$). A reduction was observed for the intermediate time period, however this finding was not statistically significant. Comparison of the elastic modulus between anterior and posterior segments showed statistically significant difference among samples provided on day 14 of use (posterior segments [814.68 ± 118.64 MPa] versus anterior segments [609.87 ± 85.79 MPa], $p = 0.003$).

Regarding the UTS, the maximum values were measured at day 0 of use and the lowest at day 14 (Figure 2). Statistically significant decreases were measured for average UTS between day 0 (35.12 ± 3.83 MPa) and day 7 (30.85 ± 6.46 MPa, $p = 0.01$), and from day 7 to day 10 (23.43 ± 11.64 MPa, $p = 0.05$). No statistically significant differences were detected between anterior and posterior samples on any period assessed.

Assessment of the yield stress (Figure 3) showed a statistically significant reduction in the average yield stress during clinical use from day 0 of use (33.05 ± 3.76 MPa) to day 7 (28.90 ± 6.07 MPa, $p = 0.0018$). Anterior and posterior segments measurements did not show a statistically significant difference.

Samples hardness evaluation showed, a decrease in the average hardness between day 0 and day 14 of use, which proved not statistically significant (Figure 4). It was also found that the posterior segments (measured at the first molars) exhibited statistically significantly higher values of Shore D hardness in comparison to anterior segments (measured at central incisors) in all periods of treatment ($p < 0.01$). Among the values measured for the anterior segments, statistically significant differences were observed from day 0 of use (57 ± 2 Shore D) to day 10 (54.38 ± 1.60 Shore D). The posterior segments demonstrated a statistically significant reduction in hardness from day 0 of use (64.06 ± 1.91 Shore D) to day 7 (61.38 ± 1.70 Shore D, $p < 0.01$).

In regard to the measurements obtained for the unprocessed material samples prepared with a sharp tool, the mean elastic modulus was calculated at 1269.53 ± 131.56 MPa, the mean UTS at 51.76 ± 2.83 MPa, the mean yield stress at 50.87 ± 2.92 MPa

TABLE 1 | Tensile testing results on aligner samples of days 0, 7, 10, 14 (mean value ± SD) and *p*-values.

		Intraoral use (days)					
Mechanical property	**Segment**	**0**	**7**	**10**	**14**	**Days**	***p***
Elastic modulus (MPa)	Average of all segments	911.66 ± 139.90	825.98 ± 111.06	819.25 ± 175.37	746.41 ± 145.41	0–7	0.114
	Anterior	941.70 ± 195.69	852.54 ± 102.25	821.54 ± 262.58	609.87 ± 85.79	7–10	0.787
	Posterior	895.49 ± 105.02	811.68 ± 116.91	817.87 ± 113.80	814.68 ± 118.64	10–14	0.246
	p-value (anterior–posterior)	0.93	0.34	0.96	0.003**	0–14	0.038*
Ultimate tensile stress (MPa)	Average of all segments	35.12 ± 3.83	30.85 ± 6.46	23.43 ± 11.64	21.29 ± 10.79	0–7	0.0015**
	Anterior	33.19 ± 4.06	30.03 ± 2.48	25.57 ± 9.27	18.89 ± 10.70	7–10	0.047*
	Posterior	36.17 ± 3.41	31.29 ± 7.91	22.14 ± 13.15	22.49 ± 11.03	10–14	0.703
	p-value (anterior–posterior)	0.096	0.936	0.303	0.246	0–14	< 0.00001***
Yield stress (MPa)	Average of all segments	33.05 ± 3.76	28.90 ± 6.07	21.88 ± 10.98	19.94 ± 10.19	0–7	0.0018**
	Anterior	31.25 ± 4.00	27.94 ± 1.97	23.93 ± 8.90	17.79 ± 9.94	7–10	0.059
	Posterior	34.02 ± 3.39	29.41 ± 7.45	20.65 ± 12.34	21.01 ± 10.51	10–14	0.771
	p-value (anterior–posterior)	0.114	0.936	0.303	0.28	0–14	< 0.00001***
Shore D hardness	Average of all segments	61.71 ± 3.90	59.25 ± 4.48	58.79 ± 3.53	58.71 ± 3.83	0–7	0.06(ant.)/0.0008***(post.)
	Anterior	57 ± 2	55 ± 1.60	54.38 ± 1.60	54.25 ± 1.98	7–10	0.496/0.56
	Posterior	64.06 ± 1.91	61.38 ± 1.70	61 ± 1.51	60.94 ± 2.17	10–14	1/0.88
	p-value (anterior–posterior)	0.0001***	0.0001***	0.0001***	0.0001***	0–14	0.027**/0.0006***

*$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$.

TABLE 2 | Tensile testing results on samples of the unprocessed material (PETG) (mean value ± SD).

Preparation method for unprocessed material	Mechanical properties			
	Elastic modulus (MPa)	Ultimate tensile stress (MPa)	Yield stress (MPa)	Shore D hardness
Sharp tool	1269.53 ± 131.56	51.76 ± 2.83	50.87 ± 2.92	60.33 ± 1.03
Water jet machine	1186.82 ± 127.60	51.89 ± 1.01	49.44 ± 0.54	60.3 ± 0.95

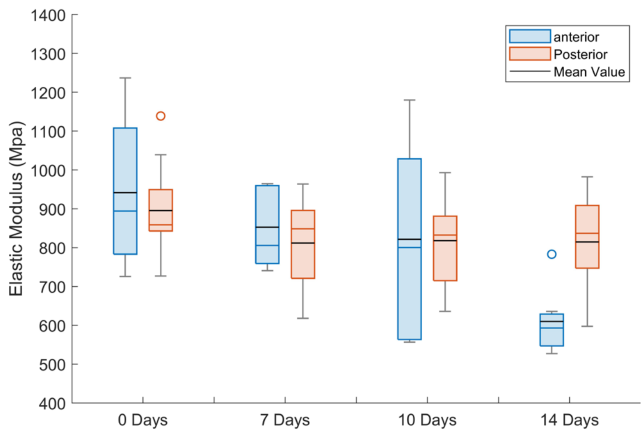

FIGURE 1 | Elastic modulus versus intraoral stay: anterior, posterior.

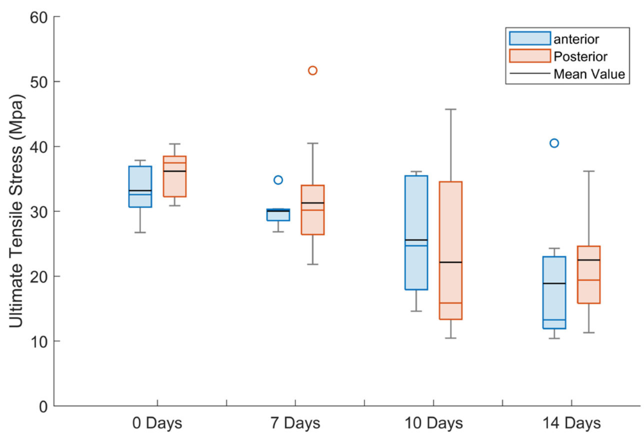

FIGURE 2 | Ultimate tensile stress versus intraoral stay: anterior, posterior.

and mean hardness at 60.33 ± 1.03 Shore D. These values were statistically significantly different from the corresponding values of the as-received aligners (day 0 of use, $p < 0.01$). For the specimens prepared with the water jet machine, mean elastic modulus was measured at 1186.82 ± 127.60 MPa, mean ultimate tensile stress at 51.89 ± 1.01 MPa, mean yield stress at 49.44 ± 0.54 MPa and mean hardness at 60.3 ± 0.95 Shore D. These values were statistically significantly different from the corresponding values of the as-received aligners (day 0 of use, $p < 0.01$), but were similar to the ones obtained from the specimens prepared with the sharp tool ($p > 0.05$).

4 | Discussion

Aligners are becoming one of the most common methods of orthodontic treatment, preferred by patients mainly due to the aesthetic superiority, faster treatment and fewer emergencies compared with fixed orthodontic appliances [32]. The daily use of aligners alters their mechanical properties; however, this alteration's effect on the clinical outcome of the orthodontic treatment is currently not known [8]. In the present study, we investigated the mechanical properties of the orthodontic aligners—Clear Aligner—after intraoral use. Although other aligners have been studied, in the literature, there is paucity of data regarding the integrity of Clear Aligner over time. This is one of the first studies that examined this product, and the first in vivo study, and provides evidence of significant alteration of its mechanical properties over time during intraoral use.

The results of our study indicate that the tensile properties and hardness of the aligners were decreased after intraoral use and therefore, the null hypothesis must be rejected. In particular, the elastic modulus was significantly decreased from day 0 of use to day 14, whereas yield stress and UTS were significantly decreased from day 0 of use to day 7. When we examined the anterior and posterior segments on day 14 of use, it was found that the posterior samples had significantly higher mean value of elastic modulus than the anterior, despite having similar values at day 0 of use. It is estimated that a potential reason for the observed difference are the higher occlusal forces which are exercised on the posterior segments [33] during that prolonged duration of use and have possibly turned the material more brittle.

According to the literature, ageing is expected to affect the structure and the mechanical properties of the orthodontic aligner materials to some extent. More specifically, in their study Schuster et al. [6] who examined patients with Invisalign aligners, found that within 2 weeks of use, aligners may present increased hardness, calcification and general wear. Bradley et al. (2016) tested the mechanical properties of Invisalign aligners used for a mean period of 44 ± 15 days and concluded that they were adversely affected by intraoral ageing. In their study, the indentation modulus (E_{IT}) in used aligners, was found to be significantly decreased compared with day 0 of use, whereas the elastic index value (η_{IT}) was increased, indicating that the aged material had turned more brittle [8]. In another study Papadopoulou et al. (2019) on Invisalign aligners after day 0, day 7 and day 14 of use, found that the aligners on day 7 and day 14 exhibited significantly inferior mechanical properties compared with day 0. However, no difference was observed between aligners on day 7 and day 14 of use [34]. It is believed that the observed effects of intraoral ageing are due to several factors, such as relief of residual stresses induced by the manufacturing process, leaching of matrix plasticizers and polyurethane softening

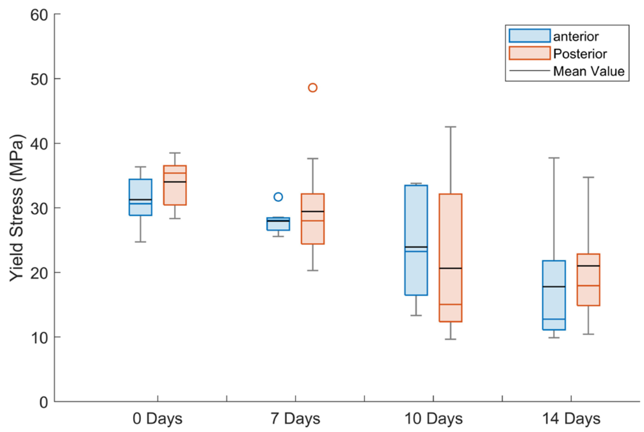

FIGURE 3 | Yield stress versus intraoral stay: anterior, posterior.

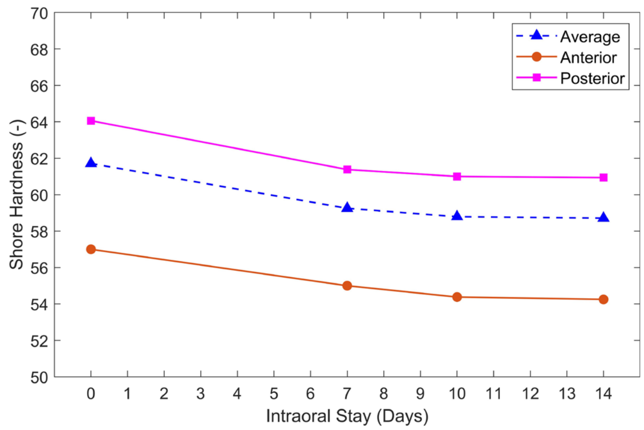

FIGURE 4 | Shore hardness versus intraoral stay: anterior, posterior, total sample.

mechanism [35]. On the other hand, Fang et al. (2020) [10] examined the mechanical properties of Invisalign, on day 0 and day 14 of use and concluded that although surface morphology of the aligner may present wear and tear after use, its mechanical properties (elastic modulus) remained stable. However, they provided evidence that trace elements, such as aluminium, can be released during clinical use, which may be a consideration, especially for patients with allergy.

Regarding Clear Aligner, Ihssen et al. (2019) studied the properties of the material used for aligner manufacturing in vitro. In their study, PETG was subjected to accelerated ageing by thermocycling after previous immersion in distilled water to simulate 14 days of intraoral use in two different temperatures, 22°C and 37°C. The estimated values for the unprocessed material ranged normally between 1.84 and 2.13 GPa for the elastic modulus, 32.2 and 41.51 MPa for the yield strength, and 40.75 and 50.26 MPa for the ultimate tensile strength. On the samples subjected to simulation of 14 days of intraoral stay, the corresponding values were 1.74–2.04 GPa for the elastic modulus, 31.93–39.80 MPa for the yield stress and 38.89–51.58 MPa for the UTS. The researchers concluded that the material's elastic modulus, yield strength and UTS were reduced compared with unused PETG specimens that were submitted to immersion. The specimens submitted to thermocycling showed a similar alteration, except for ultimate stress, which was increased [36].

These changes could negatively impact treatment outcome as the lower elastic modulus may result to decreased orthodontic forces. However, at the same time, it could benefit treatment by exerting forces on a greater deflection range and thus decreasing the risk of root resorption [37].

The results of our in vivo study are generally in agreement with those produced by Ihssen et al. (2019), although the magnitude of difference in properties of unprocessed material and after aligner intraoral ageing differ. In particular, the results of our research indicate for the unprocessed material prepared with water jet machine a mean value of 1186.82 ± 127.60 MPa elastic modulus, 51.89 ± 1.01 MPa for the UTS and 49.44 ± 0.54 MPa for the yield stress. For day 14 of intraoral ageing, the corresponding values calculated were 746.41 ± 145.41 MPa for elastic modulus, 21.29 ± 10.79 MPa for UTS and 19.94 ± 10.19 MPa for yield stress. It is hypothesised that this disparity could partly result from the different strain rate as in the present study the samples were stretched at a rate of 15 mm/min, whereas a rate of 300 mm/min was used in the other study. Further differences could stem from the material ageing process, due to different experiment design, as an in vivo study was performed in the present study, whereas for Ihssen et al. [36] ageing was conducted in vitro. In addition, other studies have shown that thermoforming may negatively impact the mechanical properties of the aligners and may have a more dominant role than ageing alone [38, 39].

This is the first study that examines hardness in Clear Aligner aligners, and we generally observed a slight decrease with ageing. This decrease may translate to diminished wear resistance, thus making the aligner vulnerable to mastication forces [9]. In particular, the posterior segments exhibited higher hardness values than the anterior ones at day 0 until day 14 of use. This may be related to the material morphology and the greater occlusal forces seen on molars in comparison to incisors [33]. This difference in forces may lead to structural modifications resulting in increased plasticity and hardness [7]. Bradley et al. (2016) [8] found similar results with decreased hardness (measured by Martens Hardness) in Invisalign aligners used for a mean period of 44 ± 15 days, in comparison with unused. Similarly, another study concluded that Invisalign samples had significantly lower hardness (measured by Martens Hardness) after day 7 and day 14 of use compared with the unused one [34]. Can et al. [11] studied 3D printed aligners after 7 days of use and also demonstrated decreased, yet not statistically different hardness (measured by Martens Hardness) compared with unused aligners. Additional studies have confirmed decreased hardness after thermoforming and ageing, noting that changes in hardness can indicate variations in applied forces that may subsequently impact the efficacy of the aligner therapy [38]. On the other hand, Schuster et al. [6] found increased hardness (measured by Vickers Hardness) in Invisalign aligners after 14 days of use and attributed the result to the alteration of the polymer crystallinity due to cold work produced by the masticatory loads.

Regarding the UTS and yield stress, in our study we found large standard deviation in the measurements at day 10 (SD: 11.64 [UTS], 10.98 [YS]) and day 14 (SD: 10.79 [UTS], 10.19 [YS]) of use. We believe that the observed differences between the tested samples are related to the increased intraoral stay and to staining and stress which turn the material brittle. In addition, imperfections may have been created by the long intraoral stay leading to earlier failures in some tests and increasing variation between the samples. Indication of the previous was found during the sample preparation stage as some samples broke after application of minimal force by the cutting tool. The latter was only observed for samples from longer intraoral stay aligners. The absence of any statistical difference between the values of unprocessed material samples that were prepared with a water jet machine and the samples prepared with the sharp tool, indicates that the sharp tool that was used for preparation of the aligners' segments should not compromise sample integrity. Further to the previous, and as a general comment regarding tensile testing, we should mention that due to the uneven morphology of the thermoformed material, matching the morphology of teeth, there is inherent non-uniformity in terms of internal stress distribution and concentration. Thus, during the tensile testing some areas, such as the interdental areas, may be subjected to locally increased stresses because of reduced material area [40] and intricate geometry with edges and curves, which can result to stress concentration. It is also noted that the thermoforming process has been confirmed to lead in uneven material thickness of the produced aligners [26]. As the mean stress was measured, the numerical data presented may be in a degree underestimated, and the results should be evaluated qualitatively.

This is the first research on orthodontic aligners that studies simultaneously multiple ageing periods and incorporates aligners that have been used intraorally, and also includes tests of samples produced by the unprocessed material. Another factor of originality is the separate study of the anterior and posterior segments of aligners. The results of our study indicate that the tensile properties of Clear Aligners potentially degrade after use, and statistically significant changes are observed after aligners' use. Greater deviations, however, are detected on the properties of the unprocessed material and the unused aligners (day 0), which indicates that the thermoforming process has significant effect on the properties of the final product. Based on the results of our study, thermoforming and ageing affect the mechanical properties of aligners' material. As the interest in orthodontic aligners is growing, it is deemed that further research on the ageing of aligner materials and their stress relaxation behaviour within the oral environment is warranted for optimal treatment design.

5 | Conclusions

The findings of our study indicate that thermoforming and ageing affect the mechanical properties of aligners. The unprocessed material displays statistically significant differences in every property tested in comparison to the material extracted from thermoformed aligners. These properties present further deterioration during clinical use on the basis of tests conducted for up to 14 days of intraoral stay.

Ethics Statement

The study was approved by the ethics committee of the National and Kapodistrian University of Athens (Athens, Greece) (protocol number: 39527).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data underlying this article will be shared upon reasonable request to the corresponding author.

References

1. Y. Ke, Y. Zhu, and M. Zhu, "A Comparison of Treatment Effectiveness Between Clear Aligner and Fixed Appliance Therapies," *BMC Oral Health* 19, no. 1 (2019): 24, https://doi.org/10.1186/s12903-018-0695-z.

2. K. Fujiyama, T. Honjo, M. Suzuki, S. Matsuoka, and T. Deguchi, "Analysis of Pain Level in Cases Treated With Invisalign Aligner: Comparison With Fixed Edgewise Appliance Therapy," *Progress in Orthodontics* 15, no. 1 (2014): 64, https://doi.org/10.1186/s40510-014-0064-7.

3. P. C. Cardoso, D. G. Espinosa, P. Mecenas, C. Flores-Mir, and D. Normando, "Pain Level Between Clear Aligners and Fixed Appliances: A Systematic Review," *Progress in Orthodontics* 21, no. 1 (2020): 3, https://doi.org/10.1186/s40510-019-0303-z.

4. L. Robertson, H. Kaur, N. C. F. Fagundes, D. Romanyk, P. Major, and M. C. Flores, "Effectiveness of Clear Aligner Therapy for Orthodontic Treatment: A Systematic Review," *Orthodontics & Craniofacial Research* 23, no. 2 (2020): 133–142, https://doi.org/10.1111/ocr.12353.

5. G. Rossini, S. Parrini, T. Castroflorio, A. Deregibus, and C. L. Debernardi, "Efficacy of Clear Aligners in Controlling Orthodontic Tooth Movement: A Systematic Review," *Angle Orthodontist* 85, no. 5 (2015): 881–889, https://doi.org/10.2319/061614-436.1.

6. S. Schuster, G. Eliades, S. Zinelis, T. Eliades, and T. G. Bradley, "Structural Conformation and Leaching From In Vitro Aged and Retrieved Invisalign Appliances," *American Journal of Orthodontics and Dentofacial Orthopedics* 126, no. 6 (2004): 725–728, https://doi.org/10.1016/j.ajodo.2004.04.021.

7. R. Condo', L. Pazzini, L. Cerroni, et al., "Mechanical Properties of "Two Generations" of Teeth Aligners: Change Analysis During Oral Permanence," *Dental Materials Journal* 37, no. 5 (2018): 835–842, https://doi.org/10.4012/dmj.2017-323.

8. T. Gerard Bradley, L. Teske, G. Eliades, S. Zinelis, and T. Eliades, "Do the Mechanical and Chemical Properties of Invisalign^{TM} Appliances Change After Use? A Retrieval Analysis," *European Journal of Orthodontics* 38, no. 1 (2016): 27–31, https://doi.org/10.1093/ejo/cjv003.

9. A. Alexandropoulos, Y. S. Al Jabbari, S. Zinelis, and T. Eliades, "Chemical and Mechanical Characteristics of Contemporary Thermoplastic Orthodontic Materials," *Australian Orthodontic Journal* 31, no. 2 (2015): 165–170.

10. D. Fang, F. Li, Y. Zhang, Y. Bai, and B. M. Wu, "Changes in Mechanical Properties, Surface Morphology, Structure, and Composition of Invisalign Material in the Oral Environment," *American Journal of Orthodontics and Dentofacial Orthopedics* 157, no. 6 (2020): 745–753, https://doi.org/10.1016/j.ajodo.2019.05.023.

11. E. Can, N. Panayi, G. Polychronis, et al., "In-House 3D-Printed Aligners: Effect of In Vivo Ageing on Mechanical Properties," *European Journal of Orthodontics* 44, no. 1 (2022): 51–55, https://doi.org/10.1093/ejo/cjab022.

12. P. Jindal, M. Juneja, F. L. Siena, D. Bajaj, and P. Breedon, "Mechanical and Geometric Properties of Thermoformed and 3D Printed Clear Dental Aligners," *American Journal of Orthodontics and Dentofacial Orthopedics* 156, no. 5 (2019): 694–701, https://doi.org/10.1016/j.ajodo.2019.05.012.

13. T. Eliades and C. Bourauel, "Intraoral Aging of Orthodontic Materials: The Picture We Miss and Its Clinical Relevance," *American Journal*

of Orthodontics and Dentofacial Orthopedics 127, no. 4 (2005): 403–412, https://doi.org/10.1016/j.ajodo.2004.09.015.

14. N. Kohda, M. Iijima, T. Muguruma, W. A. Brantley, K. S. Ahluwalia, and I. Mizoguchi, "Effects of Mechanical Properties of Thermoplastic Materials on the Initial Force of Thermoplastic Appliances," *Angle Orthodontist* 83, no. 3 (2013): 476–483, https://doi.org/10.2319/052512-432.1.

15. M. Palinkas, C. Bataglion, C. G. de Luca, et al., "Impact of Sleep Bruxism on Masseter and Temporalis Muscles and Bite Force," *Cranio* 34, no. 5 (2016): 309–315, https://doi.org/10.1080/08869634.2015.1106811.

16. B. G. Magalhães, J. L. M. Freitas, A. C. D. S. Barbosa, et al., "Temporomandibular Disorder: Otologic Implications and Its Relationship to Sleep Bruxism," *Brazilian Journal of Otorhinolaryngology* 84, no. 5 (2018): 614–619, https://doi.org/10.1016/j.bjorl.2017.07.010.

17. W. Hahn, B. Engelke, K. Jung, et al., "The Influence of Occlusal Forces on Force Delivery Properties of Aligners During Rotation of an Upper Central Incisor," *Angle Orthodontist* 81, no. 6 (2011): 1057–1063, https://doi.org/10.2319/013111-62.1.

18. K. Nishigawa, E. Bando, and M. Nakano, "Quantitative Study of Bite Force During Sleep Associated Bruxism," *Journal of Oral Rehabilitation* 28, no. 5 (2001): 485–491, https://doi.org/10.1046/j.1365-2842.2001.00692.x.

19. T. Kosaka, M. Kida, M. Kikui, et al., "Factors Influencing the Changes in Masticatory Performance: The Suita Study," *JDR Clinical & Translational Research* 3, no. 4 (2018): 405–412, https://doi.org/10.1177/2380084418785863.

20. M. Palinkas, T. F. Borges, M. T. Junior, et al., "Alterations in Masticatory Cycle Efficiency and Bite Force in Individuals With Periodontitis," *International Journal of Health Sciences* 13, no. 1 (2019): 25–29.

21. H. Matoba, H. Kanayama, T. Kato, et al., "Temporal Change in the Occlusal Vertical Dimension and Its Involvement in Modulation of Jaw Movement in Bite-Reduced Animals," *Journal of Oral Science* 60, no. 2 (2018): 170–176, https://doi.org/10.2334/josnusd.17-0122.

22. Y. Fan, X. Shu, K. C. M. Leung, and E. C. M. Lo, "Associations of General Health Conditions With Masticatory Performance and Maximum Bite Force in Older Adults: A Systematic Review of Cross-Sectional Studies," *Journal of Dentistry* 123 (2022): 104186, https://doi.org/10.1016/j.jdent.2022.104186.

23. S. Azogui-Lévy, R. Dray-Spira, S. Attal, A. Hartemann, F. Anagnostou, and J. Azerad, "Factors Associated With Oral Health-Related Quality of Life in Patients With Diabetes," *Australian Dental Journal* 63, no. 2 (2018): 163–169, https://doi.org/10.1111/adj.12577.

24. M. S. Chambers, "Sjögren's syndrome," *ORL-Head and Neck Nursing* 22, no. 4 (2004): 22–33.

25. J. S. Kwon, Y. K. Lee, B. S. Lim, and Y. K. Lim, "Force Delivery Properties of Thermoplastic Orthodontic Materials," *American Journal of Orthodontics and Dentofacial Orthopedics* 133, no. 2 (2008): 228–234, https://doi.org/10.1016/j.ajodo.2006.03.034.

26. R. Bucci, R. Rongo, C. Levatè, et al., "Thickness of Orthodontic Clear Aligners After Thermoforming and After 10 Days of Intraoral Exposure: A Prospective Clinical Study," *Progress in Orthodontics* 20, no. 1 (2019): 36, https://doi.org/10.1186/s40510-019-0289-6.

27. International Organization for Standardization (ISO), "Metallic Materials—Tensile Testing—Part 1: Method of Test at Room Temperature," International Organization for Standardization, ISO 6892-1:2019 (Geneva: Switzerland, 2019).

28. Z. D. Jastrzebski and R. Komanduri, "The Nature and Properties of Engineering Materials," *Journal of Engineering Materials and Technology* 110, no. 3 (1988): 294.

29. K. Tanne, S. Yoshida, T. Kawata, A. Sasaki, J. Knox, and M. L. Jones, "An Evaluation of the Biomechanical Response of the Tooth and Periodontium to Orthodontic Forces in Adolescent and Adult Subjects," *British Journal of Orthodontics* 25, no. 2 (1998): 109–115, https://doi.org/10.1093/ortho/25.2.109.

30. International Organization for Standardization (ISO), "Metallic Materials—Instrumented Indentation Test for hardness and Materials Parameters," International Organization for Standardization. ISO 14577-4:2016 (Geneva: Switzerland, 2016).

31. American Society for Testing and Materials (ASTM), "Standard Test Method for Rubber Property—Durometer Hardness," ASTM International, ASTM D2240-02a:2017 (2017).

32. P. H. Buschang, S. G. Shaw, M. Ross, D. Crosby, and P. M. Campbell, "Comparative Time Efficiency of Aligner Therapy and Conventional Edgewise Braces," *Angle Orthodontist* 84, no. 3 (2014): 391–396, https://doi.org/10.2319/062113-466.

33. S. E. Johnsen, K. G. Svensson, and M. Trulsson, "Forces Applied by Anterior and Posterior Teeth and Roles of Periodontal Afferents During Hold-and-Split Tasks in Human Subjects," *Experimental Brain Research* 178, no. 1 (2007): 126–134, https://doi.org/10.1007/s00221-006-0719-9.

34. A. K. Papadopoulou, A. Cantele, G. Polychronis, S. Zinelis, and T. Eliades, "Changes in Roughness and Mechanical Properties of Invisalign® Appliances After One- and Two-Weeks Use," *Materials* 12, no. 15 (2019): 2406, https://doi.org/10.3390/ma12152406.

35. H. J. Qi and M. C. Boyce, "Stress-Strain Behavior of Thermoplastic Polyurethanes," *Mechanics of Materials* 37, no. 8 (2005): 817–839, https://doi.org/10.1016/j.mechmat.2004.08.001.

36. B. A. Ihssen, J. H. Willmann, A. Nimer, and D. Drescher, "Effect of In Vitro Aging by Water Immersion and Thermocycling on the Mechanical Properties of PETG Aligner Material," *Journal of Orofacial Orthopedics* 80, no. 6 (2019): 292–303, https://doi.org/10.1007/s00056-019-00192-8.

37. K. Vinod and D. Ze'ev, *Biological Mechanisms of Tooth Movements*, 2nd ed. (Chichester, UK: John Wiley and Sons, 2015), https://doi.org/10.1002/9781118916148.

38. K. Dalaie, S. M. Fatemi, and S. Ghaffari, "Dynamic Mechanical and Thermal Properties of Clear Aligners After Thermoforming and Aging," *Progress in Orthodontics* 22, no. 1 (2021): 15, https://doi.org/10.1186/s40510-021-00362-8.

39. F. Tamburrino, V. D'Antò, R. Bucci, G. Alessandri-Bonetti, S. Barone, and A. V. Razionale, "Mechanical Properties of Thermoplastic Polymers for Aligner Manufacturing In Vitro Study," *Dental Journal* 8, no. 2 (2020): 47, https://doi.org/10.3390/dj8020047.

40. R. Reddy, M. K. B. Jagadeeshgouda, and M. S. Malagi, "Analysis of Stress Distribution in a Curved Structure Using Photoelastic and Finite Element Method," *Journal of Mechanical and Civil Engineering (IOSR-JMCE)* 12, no. 1 (2005): 112–116, https://doi.org/10.9790/1684-1213112116.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.